

TrES-5b

R-1924

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160708 · created 2026-08-06T01:02:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457577.6918 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.05
Transit depth (Rp/Rs)^2	1.9 +/- 1.37 [%]
Orbital Inclination (inc)	89.39 +/- 2.88
Airmass coefficient 1 (a1)	1.059 +/- 0.026
Airmass coefficient 2 (a2)	-0.045 +/- 0.043
Scatter in the residuals of the lightcurve fit is	2.5 %
Best Comparison Star	#3 - [130, 445]
Optimal Aperture	3.31
Optimal Annulus	11.54
Transit Duration (day)	0.0673 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.05 vs 0.142 (deviation 0.06σ)
Duration fit vs published	0.0673 d vs 0.0758 d (deviation 1.76σ)
Mid-time O–C	4.9 min
Depth SNR	2.2
Residual scatter	2.5 %

Light curve

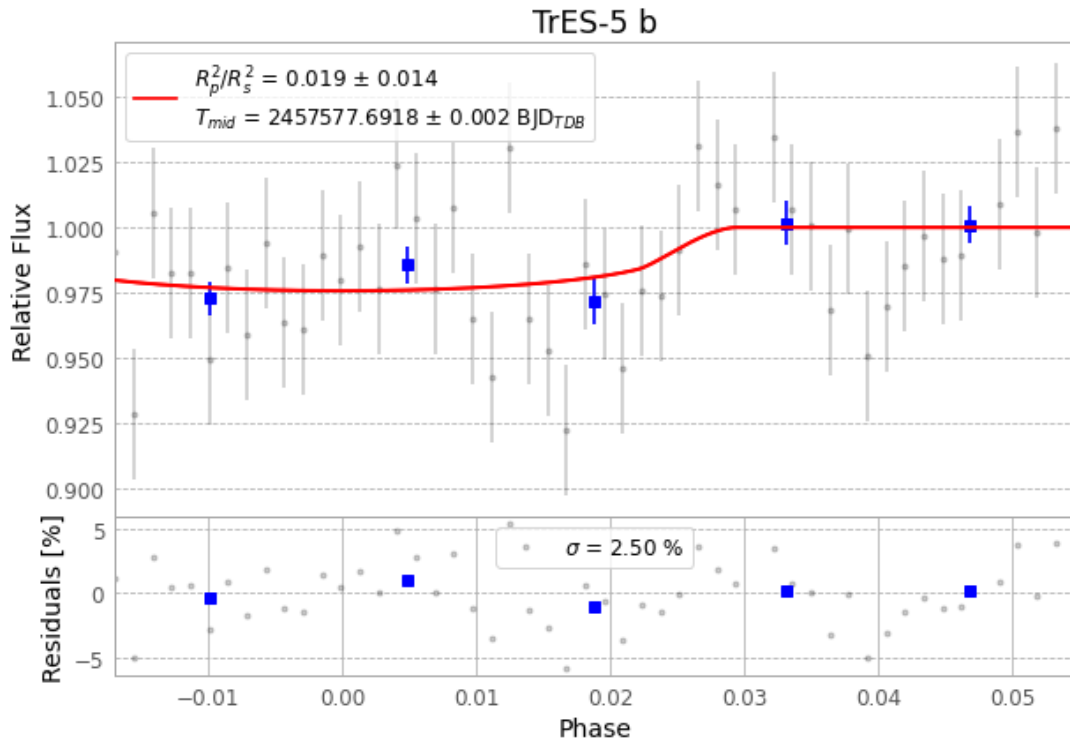


Figure 1 — FinalLightCurve_TrES-5 b_2016-07-07.png (SHA-256 a00e45bb27aff38...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 22081131c89ccabd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), TRES-5160708035712.FITS → TRES-5160708063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160708.log (e7ea697cd670...), FinalLightCurve_TrES-5 b_2016-07-07.png (a00e45bb27af...), FinalLightCurve_TrES-5 b_2016-07-07.csv (1cb9b9268a29...)

Compute resources

Wall time 107.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:02Z.